Hand Gesture Recognition Using sEMG, Time-Frequency Analysis and Artificial Intelligence

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***Abstract*—This study addresses the challenge of improving prosthetic control through the analysis of electromyographic (sEMG) signals and gesture recognition. Using the Ninapro database [1] containing EMG signals from 10 subjects performing 17 different movements, focusing on recognizing 12 specific gestures. Each gesture was repeated six times per subject, resulting in a robust dataset for analysis. The methodology employed a combination of statistical techniques, arithmetic operations and Fourier transformations to preprocess and analyze the EMG signals. Various machine learning (ML) models were evaluated, with the RandomForest classifier achieving the highest accuracy at 20.8%, while other classifiers like KNeighbors, GaussianNB, SVM(Support Vector Machines), and MLP(Multi-Layer Perceptrons) showed lower performance. To improve the results, Principal Component Analysis (PCA) was applied, but it only resulted in marginal improvements. Given the limitations of traditional ML methods, the EMG signals were converted into grayscale images and applied Fast Fourier Transform (FFT) for frequency domain analysis. A convolutional neural network (CNN) was trained on these images, achieving a high accuracy of 95% in gesture recognition. The CNN's performance underscores the potential of deep learning (DL) techniques in capturing complex patterns in sEMG signals, leading to significant improvements in real-time prosthetic control. This work contributes the foundation for future AI integration in biomedical applications, particularly in assistive technologies. Overall, this study highlights the effectiveness of artificial intelligence in assistive engineering through detailed exploration of sEMG signal processing and gesture recognition.**

***Index Terms*—sEMG, pattern recognition, neural networks (CNN), frequency analysis, artificial intelligence**

I. Introduction

Myoelectric control of prosthetic limbs has been under development since the mid-1940s. Despite advances in computing power and algorithm performance, clinically available prostheses still rely on early control strategies. The aim of this review is to detail the development, advantages, and disadvantages of prosthetic control systems, as well as to highlight areas that currently pose barriers to the transition from the lab to clinical practice [2]. While cosmetic features, weight, battery life, and prosthetic components have significantly improved over time, fundamental control strategies in clinical settings have remained largely unchanged. Amputees still control prostheses using one muscle group to open the hand and another to close it. Some advanced devices allow for wrist movements or specific grip patterns. However, these strategies fall short of matching the capabilities of the natural human hand, which can be frustrating for prosthesis users, especially in an era of constant technological innovation [2].

In contrast to early approaches that worked with one or two degrees of freedom, pattern recognition algorithms and other machine learning techniques have the potential to map more movements across multiple degrees of freedom. This is achieved by deciphering useful features present in electromyographic (sEMG) signals [3].

Automatic pattern recognition and prosthetic control are crucial research areas at the interface between humans and machines. This article delves into several key topics related to these disciplines, from the fundamentals of Machine Learning (ML) and Deep Learning (DL) to the processing of electromyographic (sEMG) signals, the creation of imagery from sEMG amplitude data and the training of Artificial Intelligence. Machine Learning and Deep Learning have revolutionized the approach to solving complex problems. These techniques enable machines to learn patterns from data and make decisions based on that information. In the context of pattern recognition, these methodologies are essential for extracting knowledge from sEMG signals. Machine Learning encompasses a wide range of algorithms and techniques that allow computers to learn from data and make predictions based on it. In this study, several ML classifiers are used to distinguish between different movement patterns based on sEMG signals. Deep Learning, a subfield of Machine Learning, involves neural networks with many layers (deep neural networks) that can automatically learn representations from raw data. This is particularly useful for recognizing intricate patterns in sEMG signals, where the high dimensionality and complexity of the data require sophisticated models to achieve high accuracy [4].

sEMG signals capture the electrical activity of muscles and are used to control prosthetics and other devices. This article explores various statistical features derived from sEMG signal amplitude. These features include Root Mean Square (RMS), Mean Absolute Value (MAV), entropy, variance, kurtosis, and symmetry. By extracting and analyzing these features, muscle movements can be effectively classified and controlled [5]. Several classifiers are considered for EMG signal classification, including methods such as Kruskal-Wallis test, Gaussian Probability Density Function (PDF), K-Nearest Neighbors (KNN), Gaussian Naive Bayes (GaussianNB), Support Vector Machine (SVM), Random Forest, and Multi-Layer Perceptron (MLP). The performance of these classifiers is evaluated using confusion matrices, which provide a detailed breakdown of accuracy and classification errors [6].

Principal Component Analysis (PCA) is employed to reduce the dimensionality of the original data. This is crucial for pattern recognition as it simplifies the data while preserving its most important characteristics. PCA transforms the data into a new set of variables (principal components) that capture the most variability in the data, facilitating its analysis and classification [7]. In addition to statistical feature analysis, the creation of imagery from EMG signal amplitude values is explored. This approach highlights significant differences between movements and enhances classification accuracy. Methods such as Fast Fourier Transform (FFT) are used to transform signals

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into the frequency domain, allowing the creation of images that capture specific frequency information. This maximizes the visual information available for movement classification, enabling more effective differentiation between various movement patterns [8]. These methods are chosen for their ability to highlight key differences in EMG signal data, thereby facilitating more precise and efficient classification.

II. Theoretical Framework

A. Machine Learning

Supervised classification algorithms are fundamental in data analysis and have various applications in classifying electromyographic (sEMG) signals. In this context, several methods can be used to identify and categorize muscle activity patterns. Below are some of the most commonly used algorithms in sEMG signal classification and their functioning.

KNeighbors (k-Nearest Neighbors): The k-NN algorithm is a supervised classifier based on the proximity between data points. It works by determining the class label of a new point according to the majority of the labels of its "k" nearest neighbors in the feature space. In the context of sEMG, k-NN can be used to classify muscle movements or detect specific patterns in sEMG signals [7].

GaussianNB (Gaussian Naive Bayes): GaussianNB is a probabilistic classifier based on Bayes' theorem. It estimates the probability that a point belongs to a given class, assuming independence between features (hence the term "naive"). This algorithm is applicable in sEMG to classify patterns of muscle activity into different categories.

SVM (Support Vector Machine): SVM is a linear and non-linear classifier that seeks to find an optimal hyperplane to separate classes. It works by maximizing the margin between classes or using kernel functions to map data to a higher-dimensional space. In sEMG classification, SVM is useful for classifying muscle movements when the classes are linearly separable or by using kernel tricks for non-linear data [9].

RandomForest: RandomForest is an ensemble of decision trees that combines multiple trees to obtain a more robust prediction and avoid overfitting. In sEMG applications, it can be used to classify complex patterns in EMG signals and detect specific movements [9].

MLP (Multilayer Perceptron): MLP is an artificial neural network with hidden layers that learns non-linear representations of data through backpropagation. This algorithm is useful for classification tasks and pattern detection in muscle signals, providing a high modeling capacity for complex sEMG data.

Confusion Matrix: Although not a classifier itself, the confusion matrix is a crucial tool for evaluating the performance of classifiers. It shows the number of true positives, false positives, true negatives, and false negatives, helping to assess the accuracy and balance between classes in a classifier. In sEMG signal analysis, this tool is essential for measuring the effectiveness of the classification algorithms used [10].

B. Fast Fourier Transform

The Fast Fourier Transform (FFT) is a mathematical technique used to convert signals between the time or spatial domain and the frequency domain. In the context of surface electromyographic (sEMG) signal processing, FFT is crucial for analyzing and representing muscle signal characteristics. The sEMG signals are captured from the skin surface using AgCl electrodes, followed by hardware filtering to enhance processing efficiency [11]. By applying FFT, the signals are decomposed into their frequency components, representing them as sums of sine and cosine functions with different frequencies, amplitudes, and phases. This frequency domain analysis helps identify dominant frequencies associated with different muscle movements, providing a clearer understanding of the underlying patterns in sEMG signals [12].

Furthermore, converting sEMG signals into images using FFT involves organizing the data into a matrix formed from 8 rows of 200-sample windows with a 100-sample overlap. Additional rows are generated through operations performed on the initial 8 rows. This matrix-based approach allows for a visual representation of the recorded electrical activity, facilitating the identification and analysis of complex muscle activation patterns that may not be evident in raw data [9]. By transforming the signals into images, advanced techniques such as Principal Component Analysis (PCA) and artificial neural networks can be applied, enhancing the accuracy of pattern recognition in applications like prosthetics, rehabilitation, and robotic control. This image-based processing is not only beneficial for diagnosing muscle diseases and tracking rehabilitation progress but also improves the effectiveness of machine learning algorithms in predictive modeling.

C. Deep Learning

Deep learning, a subfield of machine learning, involves neural networks with many layers (deep neural networks) that can automatically learn representations from raw data. This approach excels in capturing complex patterns in data, making it particularly useful for tasks such as image and speech recognition, natural language processing, and, in this case, sEMG signal processing. Convolutional Neural Networks (CNNs) are a type of deep learning model specifically designed to process data with a grid-like topology, such as images. By applying filters to the input data, CNNs can detect spatial hierarchies of features, leading to robust and accurate classification of sEMG signals. In this study, CNNs were utilized to transform sEMG signals into grayscale images, achieving high accuracy in gesture recognition and showcasing the potential of deep learning in enhancing prosthetic control systems [13]. Recent studies have demonstrated the effectiveness of deep learning in sEMG-based gesture recognition. For instance, a study by Zanghieri (2023) highlighted the use of a 1D-CNN architecture to address variability in sEMG signals, achieving significant improvements in inter-session and inter-day generalization [14]. Additionally, Tsinganos et al. (2018) discussed the application of deep learning methods to electromyography-based gesture recognition, emphasizing the advantages of CNNs in capturing the temporal and spatial features of sEMG signals [15].

III. Methodology

The procedure for this project involved the automatic recognition of patterns to detect various gestures using electromyographic (sEMG) signals. Below is a detailed description of each step taken during the research.

A. DataSet Acquisition

Acquisition Protocol

The dataset includes 52 different movements, but the focus of this work was specifically on 12 finger movements. Subjects performed these movements, following visual cues displayed on a laptop, with each movement lasting about 5 seconds followed by a 3-second rest. The upper Myo bracelet is positioned near the elbow, with the first sensor located at the radiohumeral joint, following the standard Ninapro setup for equally spaced electrodes. The second Myo cuff is placed just below the first, closer to the hand, and tilted at a 22.5-degree angle. This arrangement ensures comprehensive and consistent muscle mapping while remaining cost-effective. The Myo sensors do not

require arm shaving, and the cuff adjusts to fit snugly on the subject's arm within minutes(see figure 1). The experiment is organized into three sections: basic finger movements, hand configurations and wrist movements, and grasping or functional actions [1].

Dataset Variables

Each MATLAB file in the dataset contains synchronized data for each exercise and subject. Key variables include the subject number, sensor type, and sampling frequency. The sEMG signal data spans 16 columns, with the first eight columns from electrodes positioned around the forearm and the remaining eight from a second Myo sensor. Additional data includes three-axis accelerometer readings, cyberglove sensor data indicating joint angles, and movement labels (both original and refined). The dataset also records repetitions of movements, along with subject details such as age, gender, weight, height, handedness, and forearm circumference [1].

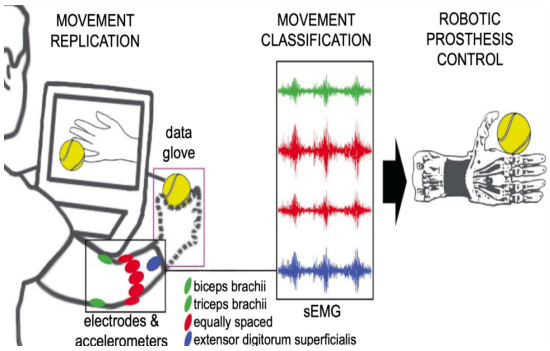

Figure 1. Electrode Placement on the Arm for Data Acquisition [1]

B. Creation of the DataFrame

A DataFrame was constructed to organize the EMG signal data in a table. This DataFrame included six statistical features commonly used in the literature for temporal analysis of sEMG signals: RMS (Root Mean Square), MAV (Mean Absolute Value), variance, entropy, kurtosis, and symmetry. These features were selected for their relevance in evaluating muscle activity and their ability to capture various properties of sEMG signals, such as intensity, variability, and complexity.

C. Application of Classifiers

Various classifiers were implemented to recognize movement patterns from the obtained statistical features. The classifiers used were KNeighbors, GaussianNB, SVM, RandomForest, and MLP. Each of these classifiers was evaluated using the original DataFrame, exploring different supervised classification approaches to determine which could offer the best accuracy in identifying movements.

D. Conversion of sEMG Signals to Grayscale Images

The amplitude values of the sEMG signals were converted to grayscale images using windows of 200 samples with an overlap of 100 samples. Images were generated from matrices created with these windows, initially using 8 rows corresponding to channels 9 to 16, and adding more rows through different operations such as sum of squares, multiplication, and normalization from 0 to 255 for grayscale.

E. Image Processing

The Fast Fourier Transform (FFT) was applied to the rows to transform the signals into the frequency domain. This processing allowed the creation of images capturing specific frequency information, which provided an additional perspective on the sEMG signals. It was observed that FFT-generated images showed significant differences between movements. An AI was trained with a dataset of these images, divided into test, validation, and training sets. Examples of images generated through FFT for movement 5:

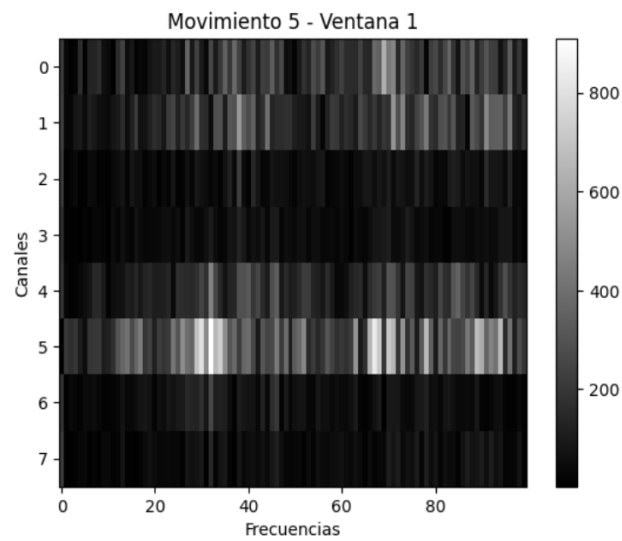

Figure 2. Representative image of the FFT applied to the 200-sample window of the sEMG signal

F. Model Definition and Training

A convolutional neural network (CNN) was used for pattern recognition in EMG signals. The network included:

- Convolutional layers with filters to detect local features.
- Max-pooling layers to reduce dimensionality and focus on important features.
- A flattening layer to transform 2D feature matrices into a 1D vector.
- A fully connected dense layer with 512 units and ReLU activation.
- A dropout layer with a 50% rate to prevent overfitting.
- An output layer with 12 units and softmax activation for class membership probabilities.

The following image shows a more detailed structure of the network structure used:

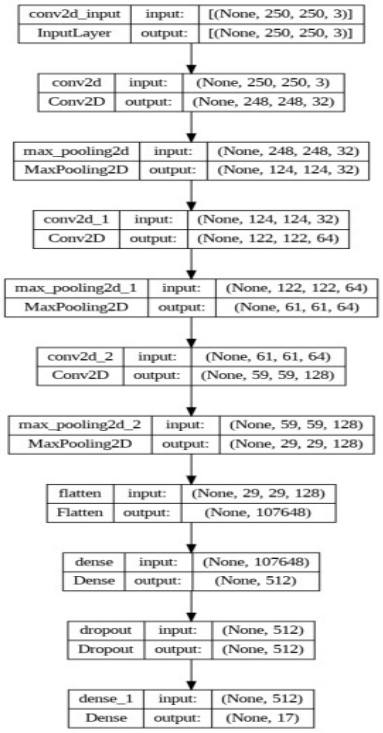

Figure 3. Neural network architecture with the best performance achieved

The model was compiled using the Adam optimizer with a learning rate adjusted for proper convergence, and the loss function used was categorical cross-entropy. The primary metric for evaluating the model's performance was accuracy. In this study, is evaluated the performance of the model using a confusion matrix, which summarizes the prediction results on a classification problem. The matrix is used to calculate important evaluation metrics such as recall, which reflects the ability of the model to correctly identify all relevant instances for each class. The recall for each class is calculated as:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

The average of these recalls, known as Macro Recall, is also computed to assess the overall performance of the model across all classes.

Additionally, the F1-Score is calculated for each class to provide a more comprehensive evaluation by balancing precision and recall. The F1-Score is particularly useful when there is an uneven class distribution or when both false positives and false negatives need to be considered. The F1-Score is calculated as follows:

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The Macro F1-Score, which is the average of the F1-Scores across all classes, offers insight into the model's overall performance.

IV. Results

A. Kruskal-Wallis Test Results

The non-parametric Kruskal-Wallis test was applied to evaluate the possibility of reducing data dimensionality as shown in table I.

Table I
Kruskal-Wallis results

Column 1	Column 2	Test statistic	p-value
RMS	Varianza	4913.97148	0.0
RMS	MAV	492.2318	4.6579e-109
RMS	Kurtosis	492.49258	0.0
RMS	Simetría	8629.4401	0.0
RMS	Entropía	821.27869	3.5909e-182
Varianza	MAV	5985.7612	0.0
Varianza	Kurtosis	7797.5195	0.0
Varianza	Simetría	8636.9835	0.0
Varianza	Entropía	6515.8997	0.0
MAV	Kurtosis	3342.4007	0.0
MAV	Simetría	874.5879	0.0
MAV	Entropía	8.7260	0.0031
Kurtosis	Simetría	1045.0233	0.0
Kurtosis	Entropía	5486.5156	0.0
Simetría	Entropía	8632.3681	0.0

The results from the Kruskal-Wallis test reveal that none of the p-values are close to 1. Instead, all p-values are significantly low, with most being exactly 0. This suggests that there are statistically significant differences between the groups being compared, leading to the rejection of the null hypothesis (H0) in all cases.

Given these results, it is evident that reducing the dimensionality of the data by eliminating any of these features would lead to a loss of critical information. Thus, all features must be retained in the subsequent analysis to ensure that the model captures all relevant patterns in the data.

B. Classification Results

The following table summarizes the accuracy of each classifier:

Table II
Accuracy Machine Learning Techniques

Classifier	Accuracy (%)
KNeighbors	19.67
GaussianNB	12.79
SVM	18.80
RandomForest	20.83
MLP	9.02

Given the unexpected results (see table II), Principal Component Analysis (PCA) was applied to reduce data dimensionality. However, the highest accuracy achieved was 19.5% with the SVM classifier. Based on these results, the decision was made to process images instead of data.

C. Model Training and Validation Performance

During the training process, both the loss and accuracy of the model were monitored for each epoch. The following graph illustrates the training and validation loss and accuracy over 25 epochs.

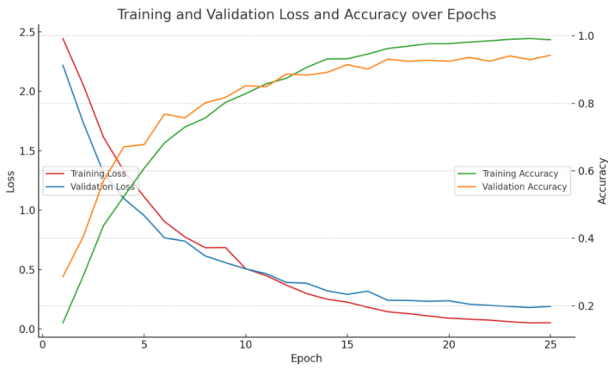

Figure 4. Training and validation performance of the CNN model over 25 epochs.

The model achieved a 95% accuracy in automatic movement recognition. Validation was performed by selecting random images and printing the AI's prediction. A confusion matrix was generated for the twelve gestures, indicating the number of images correctly recognized. The diagonal elements represent the number of correct predictions for each class, while the off-diagonal elements represent the misclassifications (see figure 5).

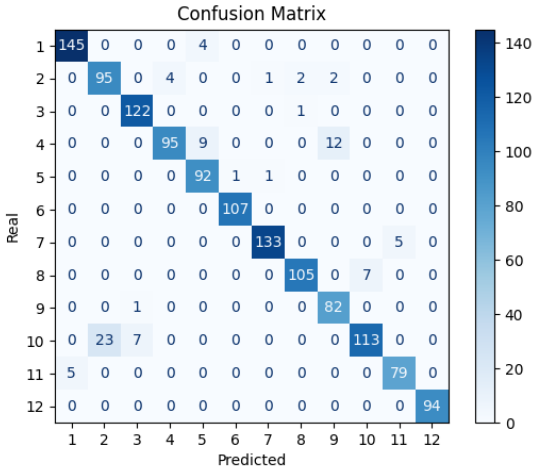

Figure 5. Confusion Matrix for 12 Gesture Classifications with Model Accuracy and Misclassifications

The matrix is used to calculate important evaluation metrics such as recall, which reflects the ability of the model to correctly identify all relevant instances for each class.

The average of these recalls, known as Macro Recall, is also computed to assess the overall performance of the model across all classes. The following table presents the recall values for each class based on the confusion matrix.

V. Discussion

The Kruskal-Wallis test was conducted to evaluate the possibility of reducing the dataset's dimensionality based on the six statistical features. The test results, as shown in Figure 1, indicate that for all feature pairs, the null hypothesis (H0) was rejected. This implies that the features are significantly different from each other, justifying the inclusion of all six features in the analysis.

Class	Recall	F1-Score
1	0.973	0.980
2	0.931	0.950
3	0.992	0.996
4	0.819	0.855
5	0.979	0.986
6	1.000	1.000
7	0.965	0.971
8	0.938	0.945
9	0.988	0.994
10	0.790	0.825
11	0.940	0.951
12	1.000	1.000
Macro	**0.976**	**0.974**

Table III
Recall and F1-Score for each class

Given the poor performance of the classifiers (see table II) with the original dataset, PCA (Principal Component Analysis) was applied to reduce the dimensionality from six features to four. The highest accuracy achieved with the PCA-reduced dataset was 19.5% using the SVM classifier. This marginal improvement suggests that while PCA helps in reducing data complexity, it might also result in the loss of important information, as evidenced by the minimal increase in accuracy.

Due to the limited success with traditional classifiers, the decision was made to convert the EMG signals into grayscale images for better pattern recognition. Various arithmetic operations were tested to create these images. However, none of these operations resulted in significant differences between the images of the different movements, and eventually, FFT (Fast Fourier Transform) was applied to work in the frequency domain. This approach showed visual differences between images of different movements (see Figure 2).

Using these images, a convolutional neural network (CNN) was trained(see figure 3), achieving a 95% accuracy in automatic movement recognition. Validation involved selecting a random image and predicting the movement type, demonstrating the CNN's effectiveness in capturing complex patterns in the EMG signals

The loss and accuracy graph(see figure 4) shows that the training loss steadily decreased over the epochs, while the training accuracy correspondingly increased, indicating that the model was learning effectively. The validation loss and accuracy also improved, albeit with some fluctuations, which is typical in neural network training due to the randomness in the validation data selection. By the end of the training process, the model achieved high accuracy and low loss on both the training and validation datasets, demonstrating good generalization ability. The final validation accuracy of 95% confirms the CNN's effectiveness in automatic movement recognition from EMG signals.

The confusion matrix shown in Figure 5 offers a clear summary of the classification model's performance. The model performs exceptionally well for most classes, as evidenced by the high values on the diagonal. Classes 1 and 2 show minimal misclassifications, and these are minor compared to the correct predictions. However, some classes, such as Class 5 and Class 11, have notable misclassifications. For instance, Class 5 has been misclassified as Class 10 in 12 instances. Class 11 also shows some misclassifications into Class 2 and Class

8, indicating areas where the model could improve in distinguishing between these classes.

In terms of recall, the model demonstrates strong performance across most classes, with an overall Macro Recall of 0.976 (see Table III). This indicates that the model is highly effective at correctly identifying relevant instances for most classes. However, the recall for Class 10 is notably lower at 0.790, suggesting that the model struggles to correctly identify instances of this class.

The F1-Score further confirms these observations. The overall Macro F1-Score is 0.974(see Table III), reflecting a good balance between precision and recall across all classes. However, as with recall, the F1-Score for Class 10 is lower, at 0.825, highlighting the same issue in correctly classifying this class. These metrics suggest that while the model is robust overall, improvements could be made to enhance the accuracy and precision for specific classes, particularly Class 10 and Class 11.

VI. Conclusion

The analysis of the dataset, which includes data for 52 different movements with a specific focus on 12 finger movements, offers critical insights into the complexity of sEMG signals and reveals significant opportunities for improving movement recognition. This meticulously designed dataset, utilizing the Myo armband with strategically placed sensors, provides comprehensive and consistent muscle coverage, which is invaluable for detailed analysis.

Applying various classifiers and dimensionality reduction techniques to this rich dataset highlights both the challenges and potential of sEMG signal analysis. Traditional classifiers, while capable of pattern recognition, demonstrated limited success due to the intricate nature of the signals. The application of Principal Component Analysis (PCA) for dimensionality reduction yielded only marginal improvements, suggesting that vital information may be lost during this process.

A notable advancement was achieved by shifting to image-based processing methods, such as grayscale images and frequency domain analysis via Fast Fourier Transform (FFT). These sophisticated techniques markedly enhanced the accuracy of movement recognition, showcasing the dataset's capability to support more advanced analysis methods.

The dataset's design, which includes synchronized sEMG signals, accelerometer readings, cyberglove sensor data, and detailed subject information, is a major asset. It not only supports a thorough evaluation of movements but also highlights the need for advanced signal processing techniques and robust neural network models to fully leverage the data's potential.

Based on these findings, several recommendations are proposed to further refine the analysis and improve classification accuracy. These include enhancing preprocessing techniques, optimizing model parameters, exploring advanced neural network architectures, and adopting comprehensive evaluation metrics. Implementing these recommendations can build on the strengths of the dataset and address current limitations, leading to more robust and accurate movement recognition from sEMG signals.

Overall, the work demonstrates that while traditional techniques have limitations, the image-based approach and advanced artificial intelligence methods, supported by the high-quality dataset, hold great promise for enhancing prosthetic control and other biomedical applications. Future research should focus on integrating advanced signal processing techniques with deep learning methods to achieve better results in movement recognition, maximizing the potential of the dataset used.

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